

Tracer Adaptive LesionLocator Ensemble

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Abstract. Interactive whole-body PET/CT segmentation must preserve a strong automatic mask while translating sparse corrections into changes of the intended lesion instances. We present a stateless system for FDG and PSMA studies that combines a five-fold residual-encoder U-Net ensemble, tracer-specific calibration, and a prompt-conditioned distance-transform model. The central contribution is a component-wise topology ledger. Candidate additions are accepted only when they do not merge accepted lesions, and supervised removals are accepted only when they do not fragment them. Input-grid annotations are mapped to cropped model space with one explicit origin subtraction, and cumulative annotations are replayed from the same image-only prediction at every call. On nine defined local interaction trajectories, the final policy increased mean AUC-Dice from 0.8403 to 0.8443 and mean AUC-DMM/F1 from 0.9136 to 0.9139, with no observed case-level regression. A five-case preliminary evaluation reported Dice 0.8546, lesion F1 0.8342, and mean metric position 3.5. Container-level checks on one FDG and one PSMA fixture produced byte-identical outputs before and after runtime modularization. These limited experiments establish behavioral reproducibility, but do not determine generalization to the hidden final set.

Keywords: autoPET V · interactive segmentation · PET/CT · topology

1 Introduction

Whole-body PET/CT combines metabolic and anatomical information, but lesion appearance and physiological uptake differ substantially between FDG and PSMA tracers. Automatic segmentation therefore faces both voxel-level uncertainty and an instance-level problem: a mask may obtain high overlap while missing a small lesion, retaining a physiological false positive, or joining separate tumors. Interactive evaluation adds a further requirement. A useful correction should modify the lesion indicated by a sparse annotation without destabilizing unrelated parts of the volume.

AutoPET V evaluates this setting through cumulative simulated and clinician-driven annotations. Our starting point is the multitracer LesionTracer ensemble [4] and the promptable LesionLocator model [5], both based on nnU-Net v2 [1]. Rather than retraining either network, we study the interface between their predictions. The research question is whether simple, explicit component

invariants can make prompt fusion safer while retaining the accuracy of the image-only ensemble.

The contributions are threefold. First, we define a stateless tracer-aware pipeline that reconstructs the initial mask and replays all cumulative annotations. Second, we introduce component-wise acceptance tests that prevent one useful proposal from concealing an unrelated merge or split. Third, we publish a modular runtime with unit, policy-equivalence, and container-level reproduction evidence. All thresholds and pretrained checkpoints are frozen for the submitted system.

2 Methods

2.1 Data

Training data No model was trained for this submission. The image-only checkpoints were trained on the public AutoPET-III data by their publishers, following the multitracers training procedure in [4]. The interactive checkpoint was trained on the public AutoPET-IV data with synthetic click generation as described in [5]. These collections derive from the public FDG and PSMA PET/CT resources described in [2, 3]. We introduce no additional training subjects, labels, or weight optimization.

Validation data Engineering validation used ten local challenge-format fixtures: six FDG and four PSMA studies, each replayed for up to five cumulative interaction steps. One FDG fixture had no defined Dice or lesion-F1 value because both the target and prediction were effectively negative for the evaluated quantity. We therefore report paired aggregate metrics over the remaining nine defined trajectories. These fixtures were repeatedly used during policy development and must not be interpreted as an independent estimate of hidden-test performance.

2.2 System overview

Let $I = (I_{CT}, I_{PET})$ denote a co-registered volume and let $P_t = (P_t^+, P_t^-)$ contain all foreground and background annotation voxels available at interaction step t . The system computes

$$K_0 = C_\tau(E(I)), \quad K_t = F(K_0, D(I, P_t), P_t; \pi_\tau), \quad (1)$$

where E is the five-fold image-only ensemble, C_τ is tracer-conditioned calibration, D is the prompt-conditioned donor model, F is component-wise reconciliation, and π_τ is the fixed FDG or PSMA interaction policy. Each call reconstructs K_0 and evaluates the full cumulative P_t ; no mask from step $t - 1$ is required. This makes identical requests deterministic with respect to saved inputs and checkpoints.

2.3 Preprocessing and coordinate mapping

CT and PET volumes are processed with the normalization, target spacing, and sliding-window configuration stored in the pinned nnU-Net plans. Predictions are returned to the CT reference grid and copy its size, spacing, origin, and direction. An annotation coordinate on the input grid is converted to cropped model space by subtracting the crop origin exactly once and then applying the crop-to-resampled scale. Coordinates with malformed dimensionality or outside the volume are rejected before model execution.

2.4 Tracer routing and initial segmentation

A frozen three-classifier vote assigns each study to the FDG or PSMA policy. The image-only mask is generated by five 3-D residual-encoder U-Net folds with mirror test-time augmentation disabled. FDG foreground probabilities are thresholded at 0.47. Components smaller than 25 voxels are removed, except when at least 25 robust components indicate high disease burden, in which case the dust threshold is relaxed to five voxels. PSMA uses threshold 0.50 and a five-voxel dust threshold. A frozen logistic component classifier then rejects PSMA candidates above a false-positive probability of 0.86. The classifier uses fixed component-level image and geometry features; it is not adapted on the evaluation fixtures.

2.5 Prompt-conditioned donor

When effective annotations are present, a fold-0 EDT LesionLocator model receives CT, PET, and all cumulative prompts and produces a binary donor mask D_t . Tumor propagation is active for every annotated FDG call. For PSMA it is activated after at least six cumulative foreground points and only when K_0 contains at most 128 connected components. This evidence threshold limits changes from very sparse prompts, while the component ceiling avoids applying the donor to extremely fragmented initial masks.

2.6 Topology ledger

All masks use 18-connectivity. Let $n(M)$ be the number of foreground connected components in mask M . The ledger starts from $M = K_0$ and is the only module permitted to mutate the accepted mask. For each donor component A selected by a foreground annotation, addition is accepted iff

$$n(M \cup A) \geq n(M). \quad (2)$$

Because proposals are evaluated one connected component at a time, Eq. 2 rejects a proposal that bridges two accepted lesions instead of allowing a separate newly discovered component to compensate for that merge. On acceptance, $M \leftarrow M \cup A$ and the component count is recomputed.

Foreground annotation strokes are also direct supervision. Each connected stroke is added without dilation and is subjected to the same non-merging test. Broad background component replacement is disabled in the submitted runtime. For PSMA, an explicitly annotated background stroke R is removed only iff

$$n(M \setminus R) \leq n(M), \quad (3)$$

which prevents a thin erasure from splitting one accepted lesion into several objects. FDG background strokes do not mutate the submitted mask. These rules preserve component-count direction, not full digital topology; this distinction is considered in the limitations.

2.7 Postprocessing and deployment

The reconciled binary mask is transposed back to the challenge array order, resampled to the reference geometry, and written as one tumor-lesion segmentation. The Grand Challenge container performs no network access during inference and reads the model archive from a read-only mount. The submitted image contains five initial folds, one interactive fold interface, the frozen tracer router, and the component-filter parameters described above.

2.8 Training and test parameters

Both segmentation networks use the pinned 3-D full-resolution nnU-Net plans: $192 \times 192 \times 192$ voxel patches, source batch size two, and a residual-encoder U-Net architecture. The upstream trainers use compound soft Dice and cross-entropy supervision. No learning rate, optimizer, scheduler, or augmentation was introduced because weights were not updated. Inference uses five-fold ensembling for K_0 , one EDT fold for interactive calls, and no mirror TTA. Engineering and validation used up to six NVIDIA H100 80 GB HBM3 GPUs. Deployment targets an NVIDIA T4 with 16 GB VRAM; the highest sampled allocation in the two container fixtures was approximately 7.1 GB.

2.9 Implementation and availability

Challenge I/O, initial prediction, donor execution, and reconciliation are separate modules. Prompt validation and tracer policy are immutable data objects, while the topology ledger owns mask changes. The Apache-2.0 source, tests, build recipes, notices, checkpoint hashes, and preprint are available at github.com/marchmello1/Tracer-Adaptive-LesionLocator-Ensemble.

3 Experimental Setup

3.1 Paired causal replay

The v2 policy and final v3 policy were compared on identical images and cumulative annotations. At each step the two policies started from the same K_0

Table 1. Paired local interaction results. Values are trajectory AUCs; the negative FDG fixture with undefined metrics is excluded.

Case	v2 Dice	v3 Dice	v2 F1	v3 F1
FDG 1	.8591	.8591	.7500	.7500
FDG 2	.8547	.8547	1.0000	1.0000
FDG 4	.8652	.8652	1.0000	1.0000
FDG 5	.7573	.7578	.7742	.7742
FDG 6	.9489	.9489	1.0000	1.0000
PSMA 1	.8011	.8014	.8871	.8898
PSMA 2	.8343	.8361	.9502	.9502
PSMA 3	.7937	.8238	1.0000	1.0000
PSMA 4	.8480	.8515	.8611	.8611
Mean	.8403	.8443	.9136	.9139

and received the same donor output; thus the comparison isolates the exact foreground-stroke rule. Dice was computed as

$$\text{Dice}(M, G) = \frac{2|M \cap G|}{|M| + |G|}. \quad (4)$$

Lesion detection used the challenge DMM/F1 evaluator. We summarize each five-step trajectory by its causal area under the metric curve and then take the unweighted mean over the nine defined cases. No confidence interval is reported because the fixtures are a small, development-selected cohort.

3.2 Packaging reproduction

The modular policy was compared with the frozen policy in 1,000 randomized mask/prompt trials. We then ran the frozen and modular containers with identical model mounts, environment, and inputs on one FDG and one PSMA fixture. Finally, the runtime was embedded into a self-contained Docker image and rerun without a host code mount. Output files were compared by SHA-256, not merely by aggregate metrics. The repository contains 45 focused tests for I/O validation, policy activation, component filtering, topology guards, idempotence, and release layout.

4 Results

Table 1 shows a mean change of +0.0040 Dice AUC and +0.0003 F1 AUC. Dice improved in five cases and was unchanged in four. F1 improved in one case and was unchanged in eight; no defined trajectory regressed. The largest Dice change occurred for PSMA 3 (+0.0301), where exact foreground supervision restored voxels that the donor alone did not retain.

The earlier frozen v2 image was evaluated on the five-case preliminary set and obtained Dice 0.854649, lesion F1/DMM 0.834179, and mean metric position 3.5. One negative case had undefined Dice and F1, so those means used four cases. The modular v3 image was not separately submitted to the preliminary set.

All 1,000 randomized production-policy comparisons agreed. The FDG and PSMA container fixtures were byte-identical between the frozen and modular runtimes, and the self-contained PSMA rerun retained the same output hash. These results establish equivalence for the tested conditions, not a formal proof over all image geometries or GPU execution paths.

5 Discussion

The results support a narrow conclusion: explicit component-level guards can improve correction behavior without changing network weights or degrading the observed development trajectories. The per-component evaluation is important. A batch union may preserve its total component count when one proposal creates a lesion while another joins two existing lesions; processing components individually prevents that cancellation. Exact foreground strokes provide a second, model-independent path for honoring supervision, while the non-merging guard limits their spatial effect.

The tracer policies reflect different failure modes. FDG preserves a strong automatic mask and accepts prompt-selected donor components immediately. PSMA uses stricter evidence and a frozen false-positive filter because physiological uptake and high lesion burden create more ambiguous components. These choices are transparent and testable, although fixed thresholds may not transfer to a different scanner or cohort distribution.

6 Limitations

The validation cohort is small, repeatedly inspected, and unsuitable for estimating final-set rank. The preliminary set contains only five studies and is likewise an implementation check. The component-count inequalities do not preserve holes, handles, or anatomical identity; they are conservative surrogates for two observed instance errors. The six-point PSMA threshold, 128-component ceiling, and probability thresholds were not estimated with an independent calibration cohort. Only one FDG and one PSMA fixture were used for container-level equivalence, and sampled memory is not a worst-case bound. Finally, frozen public checkpoints inherit the data composition, annotation quality, and subgroup biases of their original training sets. The system is a research prototype and is not intended for clinical decision-making.

7 Conclusion

Tracer Adaptive LesionLocator Ensemble combines a multitracer image-only ensemble with prompt-conditioned proposals through a stateless topology ledger.

Table 2. Algorithm and reproducibility details.

Item	Configuration
Team	Nuvo AI
Algorithm	Tracer Adaptive LesionLocator Ensemble
Input	3-D co-registered whole-body CT and FDG- or PSMA-PET
Training data	Published AutoPET-III/IV cohorts; no additional local training data
Preprocessing	Pinned nnU-Net normalization, resampling, and sliding-window inference
Architecture	3-D residual-encoder U-Net; 192 ³ patch; source batch size 2
Initial model	Five LesionTracer folds and a frozen three-model tracer vote
Interaction model	One LesionLocator EDT fold with cumulative prompt replay
Postprocessing	Tracer calibration, component filtering, and topology ledger
Loss/optimizer	Upstream Dice+cross-entropy; no submission-specific optimization
Augmentation/TTA	No new augmentation; mirror TTA disabled for submitted inference
Framework	nnU-Net v2, PyTorch, SimpleITK, and cc3d
Development hardware	Up to six NVIDIA H100 80 GB HBM3 GPUs
Inference hardware	NVIDIA T4, 16 GB VRAM

Component-wise non-merging and non-fragmenting checks make interactive changes explicit, local, and reproducible. Local paired replay suggests a small improvement without observed regression, while container tests confirm behavioral equivalence after modularization. Larger independent evaluation is required to establish generalization and clinical utility.

Acknowledgements This work uses the public AutoPET-III LesionTracer and AutoPET-IV LesionLocator projects. We thank their authors and the autoPET organizers.

Disclosure of Interests The author declares no competing interests.

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